

llmXive Automated Review of Qwen-AgentWorld: Language World Models for General Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents several quantitative claims that require tighter alignment between the text and the provided data tables. In Section 5, the text asserts a “+8.66” performance gain for the 35B model, yet Table 1 only displays results for the 397B model and the final proposed method, leaving the 35B baseline unverified in the main results table. This disconnect makes the specific gain claim difficult to validate directly from the presented evidence.

In Section 6, the manuscript claims that concurrent work “doubles performance” via auxiliary loss. However, the text does not explicitly state the baseline metric or the specific task to which this doubling applies. Without this context, the magnitude of the improvement remains ambiguous and unsupported by the immediate data.

Additionally, Section 7 compares Sim RL (50.3%) against Real RL (45.6%) but fails to clarify whether the Real RL figure corresponds to the 35B or 397B model configuration. This ambiguity undermines the precision of the “exceeds” claim. Finally, Section 4 reports a consistency range of $\rho = 0.92\text{--}0.99$ without specifying which evaluation dimensions or model pairs correspond to the lower and upper bounds, rendering the “high consistency” claim vague. These issues are primarily clarity and attribution errors that should be corrected to ensure factual accuracy.

Action items.

- **[writing]** Section 5 claims an +8.66 gain at 35B scale, but Table 1 omits the 35B baseline. Clarify the reference model for this gain.
- **[writing]** Section 6 states concurrent work ‘doubles performance’ without specifying the baseline metric. Define the baseline for this claim.
- **[writing]** Section 7 compares Sim RL (50.3%) to Real RL (45.6%) but does not specify if the baseline is the 35B or 397B model. Clarify the comparison.
- **[writing]** Section 4 cites consistency $\rho=0.92\text{--}0.99$ without breaking down by dimension. Specify which dimensions correspond to these bounds.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively visualizes the model’s architecture and performance, but the caption is marred by raw LaTeX code artifacts and missing model names. Additionally, the top chart’s x-axis label ‘Ours’ does not explicitly match the full model name in the legend.

Figure 2

The figure effectively visualizes three distinct reasoning patterns with clear text examples and summaries. However, the caption contains a likely typo or placeholder ('-397B-A17B') for the model name that should be corrected.

Figure 3

Figure 3 effectively illustrates micro-level fidelity improvements across Search, Terminal, and MCP domains as described in the caption. However, the text in the Terminal panel is rendered as a single unbroken line that obscures the character enumeration, and there is a minor spelling error in the MCP panel.

Figure 4

Figure 4 effectively illustrates the OS domain capability by showing the transition from a spreadsheet application to a print preview interface. The visual elements, including the 'Current state' and 'Simulated next state' labels, align perfectly with the caption's description of the agent clicking 'File > Print' and the model predicting the resulting backstage view.

Figure 5

Figure 5 effectively demonstrates the world model's ability to predict terminal output for a compilation task. The image clearly shows the input command and the resulting simulated output, including specific compiler warnings and the final linking step, which aligns perfectly with the caption's description.

Figure 6

The figure presents clear line plots comparing two RL strategies, but the caption is insufficient as it fails to define the 'Real RL' and 'Sim RL' groups shown in the legend, leaving the experimental setup undefined.

Figure 7

Figure 7 effectively summarizes the dataset composition and statistics, but the 'Source benchmarks' legend text is illegible due to small font size and low contrast. Additionally, the color mapping for domains in the bar charts is not explicitly defined in a legend, requiring the viewer to cross-reference other parts of the figure.

Figure 8

The figure presents a clear comparison of model performance across domains, but the caption's claim of 'five-dimensional rubric mean' is not visually supported as the dimensions are not shown. Additionally, the y-axis grouping labels are ambiguous regarding the middle subplots.

Figure 9

The figure effectively visualizes the cross-domain transfer gains described in the caption, but the inclusion of the ‘Terminal’ domain (the training domain) in the transfer plot creates conceptual ambiguity, and the specific evaluation metric is not explicitly defined.

Figure 10

The figure is a clear visual overview of the system’s domains, but the caption contains a grammatical error (missing subject) and a factual discrepancy regarding the number of categories depicted.

Figure 11

Figure 11 effectively illustrates the SWE domain capability by displaying a realistic Python traceback for an out-of-memory error during one-hot encoding. The visual content aligns perfectly with the caption’s description of the agent running a script and the world model predicting the specific error state.

Figure 12

The figure clearly illustrates the three-stage training pipeline with distinct data, objective, and technique sections for each stage. However, the caption contains a grammatical error omitting the model name, and the top-level summary text does not perfectly align with the detailed descriptions in the caption.

Action items.

- **[writing]** Figure 1: The caption contains raw LaTeX formatting artifacts (e.g., ‘=1.5pt-colordecoupleDecouple’) and missing model names (e.g., ‘Overview of .’) instead of clean text.
- **[science]** Figure 1: The top bar chart legend lists ‘Qwen-AgentWorld-397B-A17B’ as the purple bar, but the x-axis label for that bar is simply ‘Ours’, creating a disconnect between the legend and the axis.
- **[writing]** Figure 1: The bottom-left section contains a text label ‘Generalize to Claw Agent’ that appears to be a caption fragment or typo rather than a clear description of the data.
- **[writing]** Figure 2: The caption refers to ‘-397B-A17B’ (likely a placeholder or typo for the model name), which is not defined in the caption or other provided text.
- **[writing]** Figure 3: The ‘Terminal’ panel contains a rendering artifact where the text ‘t(1)e(2)s(3)...x(24)t(25)/n(26) -> 26 bytes’ is displayed as a single unbroken line, making the character enumeration and byte count calculation difficult to read.
- **[writing]** Figure 3: The ‘MCP’ panel text ‘UUID fomate’ contains a spelling error (should be ‘format’).
- **[science]** Figure 6: The legend defines ‘Real RL’ and ‘Sim RL’, but the caption fails to define

these terms or explain the experimental comparison (e.g., training on real vs. simulated data), making the figure’s scientific claim unintelligible without external context.

- **[writing]** Figure 6: The y-axis label ‘Score (%)’ is ambiguous; it is unclear if the values (e.g., 50) represent a percentage (0-100) or a raw score scaled to 100, which affects the interpretation of the F1 metric.
- **[writing]** Figure 7: The ‘Source benchmarks’ legend uses color swatches to map to domains, but the text labels (e.g., ‘MCP Mark 6’, ‘Wide Search’) are too small and low-contrast to be legible.
- **[writing]** Figure 7: The ‘Avg. context length’ and ‘Avg. trajectory turns’ charts use color-coding for domains (e.g., red for Web/Android), but the specific color-to-domain mapping is not explicitly defined in a legend, relying on the viewer to infer it from the ‘Source benchmarks’ section.
- **[science]** Figure 8: The caption claims to show ‘five-dimensional rubric mean per domain,’ but the chart displays a single score per model per domain (likely an aggregate) without showing the five dimensions or their breakdown.
- **[writing]** Figure 8: The y-axis label ‘Text-based’ and ‘GUI’ is positioned on the far left, but the chart contains seven subplots (Terminal, MCP, Search, SWE, Android, Web, OS) where only the first two are text-based and the last three are GUI; the labels do not clearly demarcate the grouping of the middle subplots.
- **[writing]** Figure 8: The x-axis labels for models are cramped and multi-line (e.g., ‘Claude Opus 4.8’), making them difficult to read; consider rotating or simplifying.
- **[science]** Figure 9: The legend lists ‘Terminal’ (red circles), but the caption states the model was trained on Terminal data alone. If Terminal is the training domain, it should not be plotted as a held-out transfer result in panel (b) alongside MCP, SWE, and Search; the inclusion of the training domain in the transfer plot is conceptually confusing or mislabeled.
- **[writing]** Figure 9: The y-axis label ‘Score (0-100)’ is present, but the specific metric (e.g., F1, Accuracy, Pass@1) is not defined in the caption or axis label, making the absolute values ambiguous.
- **[writing]** Figure 10: The caption begins with a missing subject (e.g., ‘Qwen-AgentWorld’ or ‘The model’), reading as ‘unifies seven categories...’ instead of a complete sentence.
- **[writing]** Figure 10: The caption text ‘unifies seven categories’ does not match the visual content, which depicts eight distinct categories (MCP Servers, Search Engine, IDE, Terminal/CLI, Android System, Web Browser, Operating System, and a central hub).
- **[writing]** Figure 12: The caption contains a grammatical error (‘Three-stage training pipeline of .’) where the model name is missing after the preposition ‘of’.
- **[writing]** Figure 12: The top-level title ‘CPT injects, SFT activates, RL sharpens’ is not explicitly defined in the caption, which uses slightly different verbs (‘instills’, ‘sharpens’) for the same stages.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on a dense layer of domain-specific acronyms and shorthand that significantly raises the barrier to entry for non-specialist readers. While the field of agent simulation is rapidly evolving, the paper frequently introduces terms like “LWM,” “CPT,” “SFT,” “RL,” “Sim RL,” “Real RL,” “GSPO,” and “MCP” without providing their full definitions at the point of first use.

For instance, in Section 1 (Introduction), the text states, “Training follows ‘CPT injects, SFT activates, RL sharpens.’” A reader unfamiliar with the specific jargon of this sub-field cannot parse this sentence without external knowledge. Similarly, Section 3 introduces “GSPO” as a method without defining it, and Section 5 discusses “Sim RL” vs. “Real RL” as if the distinction is self-evident.

The term “LWM” is used as a primary noun throughout the text (e.g., “LWM reasoning patterns,” “LWM warm-up”) but is never explicitly defined as “Language World Model” in the immediate vicinity of its first appearance in the main body, relying on the title for context. This creates a “black box” effect where the core subject matter is obscured by its own abbreviation.

To improve accessibility, the authors should:

1. Expand every acronym (CPT, SFT, RL, LWM, GSPO, MCP, Sim RL, Real RL) to its full form at the very first occurrence in the text.
2. Consider replacing “Sim RL” and “Real RL” with “Simulation-based RL” and “Real-environment RL” in the first instance to reduce cognitive load.
3. Ensure that “MCP” is defined as “Model Context Protocol” (or the specific full name intended) in the Introduction or the domain taxonomy table.

These changes are purely editorial and would not alter the scientific content but would make the paper significantly more readable for a broader audience.

Action items.

- **[writing]** Expand every acronym (CPT, SFT, RL, LWM, GSPO, MCP, Sim RL, Real RL) to its full form at the very first occurrence in the text.
- **[writing]** Consider replacing “Sim RL” and “Real RL” with “Simulation-based RL” and “Real-environment RL” in the first instance to reduce cognitive load.
- **[writing]** Ensure that “MCP” is defined as “Model Context Protocol” (or the specific full name intended) in the Introduction or the domain taxonomy table. These changes are purely editorial and would not alter the scientific content but would make the paper significantly more readable for a broader audience.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent logical framework for training Language World Models (LWMs) via a three-stage pipeline (CPT, SFT, RL) and demonstrates their utility in simulation and as agent foundations. The causal chain from “next-state prediction” to “improved agent performance” is well-supported by the experimental results in Table 4, where the LWM RL warm-up leads to consistent gains across diverse benchmarks. The internal logic of the “Decoupling” vs. “Unifying” paradigms is sound, and the distinction between Sim RL and Real RL is clearly maintained.

However, there are minor logical ambiguities in the phrasing of claims that could mislead the reader regarding the nature of the training and the scope of comparisons. Specifically, the claim in Section 5.2 that the method works “without fine-tuning” contradicts the explicit mention of an “LWM RL” stage, which is a form of fine-tuning. This requires a semantic clarification to “without *task-specific* fine-tuning” to maintain logical consistency. Additionally, the comparison between Sim RL and Real RL in Section 5.1 lacks an explicit statement confirming that the Real RL baseline uses the identical model architecture (35B), which is a necessary premise for the validity of the performance comparison. Finally, while the mitigation of reward hacking is described, the logical sufficiency of a 10% rule-based weight to counteract a 90% LLM judge bias is asserted but not explicitly justified with a logical argument or ablation reference in the text. Addressing these points will tighten the logical consistency of the manuscript.

Action items.

- **[writing]** The paper presents a coherent logical framework for training Language World Models (LWMs) via a three-stage pipeline (CPT, SFT, RL) and demonstrates their utility in simulation and as agent foundations. The causal chain from “next-state prediction” to “improved agent performance” is well-supported by the experimental results in Table 4, where the LWM RL warm-up leads to consistent gains across diverse benchmarks. The internal logic of the “Decoupling” vs. “Unifying” paradigms is sound, and the d

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extend beyond the immediate evidence provided, particularly regarding the generalizability and superiority of the proposed methods.

First, the claim in Section 5.1.2 that “Controllable Sim RL (50.3%) outperforms Real RL

(45.6%)” and the subsequent conclusion that simulation can “surpass real-environment training” is an overreach. While the specific numbers for WideSearch support a performance gain in that specific task, the paper extrapolates this to a general principle without sufficient discussion of the potential for overfitting to the simulated environment or the specific conditions (e.g., the nature of the “fictional worlds”) that enabled this result. The comparison is limited to one benchmark (WideSearch), yet the implication is a broad advantage of simulation over reality for agent training.

Second, the description of the transfer learning results in Section 5.2 as “Radical cross-task generalization” is hyperbolic. The paper demonstrates that a single-turn, non-agentic LWM RL warm-up improves performance on multi-turn agentic tasks across several benchmarks. While this is a positive result, the term “radical” implies a level of universality and robustness that the data does not fully substantiate. The paper does not adequately discuss the boundaries of this generalization or the specific types of tasks where this transfer might not hold, potentially misleading readers about the method’s general applicability.

Third, the statement in Section 4.2 that the model maintains “perfect consistency” in MCP schema reproduction across nine Notion API calls is likely an overstatement. Without a clear definition of “perfect” (e.g., 100% match on all fields, including dynamic ones) or statistical evidence supporting this absolute claim, it risks exaggerating the model’s reliability. A more measured phrasing, such as “high consistency” or “consistent schema reproduction,” would be more accurate given the typical challenges of maintaining state in complex API interactions.

Finally, the claim in the Introduction that \method is the “first native language world model covering seven domains” requires careful scrutiny. While the paper presents a novel integration, the existence of prior work in specific domains (e.g., web, terminal) suggests that the “first” claim might be too absolute unless the specific combination and training methodology are uniquely novel in a way that is clearly distinguished from the sum of its parts. The paper should clarify the novelty more precisely to avoid overclaiming.

Action items.

- **[science]** The claim that Sim RL ‘surpasses’ Real RL (Section 5.1.2) is overreaching. The data shows Sim RL (50.3%) outperforms Real RL (45.6%) on WideSearch F1 Item, but the paper generalizes this to a broad superiority of simulation without addressing domain specificity or the potential for overfitting to the simulated environment’s quirks.
- **[science]** The assertion of ‘Radical cross-task generalization’ (Section 5.2) overstates the evidence. While gains are reported on 7 benchmarks, the transfer is from single-turn non-agentic LWM RL to multi-turn agentic tasks. The paper does not sufficiently discuss the limitations of this transfer or the specific conditions under which it might fail, implying a universality not fully supported by the data.
- **[writing]** The claim that the model achieves ‘perfect consistency’ in MCP schema reproduction (Section 4.2) is likely an overstatement. The text states ‘perfect consistency’ across nine calls, but without explicit statistical evidence or a definition of ‘perfect’ in this context (e.g., 100% match on all fields), this phrasing risks exaggerating the model’s reliability.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a significant advancement in Language World Models (LWMs) capable of simulating complex agent environments. From a safety and ethics perspective, the work introduces novel capabilities that warrant careful scrutiny regarding dual-use potential and evaluation integrity.

Dual-Use and Misinformation Risks: The paper highlights the ability to perform “Fictional-World Construction” (Section e002, Application I) and “Controllable Simulation” (Section e002, Application I). Specifically, the authors demonstrate generating self-consistent fictional worlds (e.g., “2030 Mars migration”) and simulating specific error modes or perturbations. While framed as a tool for training agents, these capabilities directly lower the barrier for generating highly realistic, context-aware disinformation, deepfakes, or social engineering scenarios. The model’s ability to predict “realistic” search results (Section e002, Search domain) and maintain “state coherence” across turns could be exploited to create convincing fake news narratives or phishing environments. The manuscript currently lacks a dedicated “Safety and Limitations” section that explicitly addresses these dual-use risks and proposes mitigation strategies (e.g., watermarking simulated outputs, restricting access to the simulation engine).

Evaluation Integrity and Reward Hacking: The evaluation methodology relies heavily on LLM judges (e.g., GPT-5.2) to score dimensions like “Realism” and “Factuality” (Section e002, Appendix). The authors acknowledge the risk of “Reward Hacking Through Self-Praise” in Section e002, where the policy learns to insert qualitative affirmations to inflate scores. While mitigations like rule-based verifiers are mentioned, the paper does not sufficiently discuss the broader implications of using LLM judges for safety-critical evaluations. If the judge model itself has biases or if the “self-praise” mitigation is imperfect, the reported safety and fidelity metrics may be inflated, leading to a false sense of security when deploying these models in real-world agent systems.

Harmful Content in Training and Evaluation: The benchmark covers “SWE” (Software Engineering) and “Terminal” domains, involving the prediction of code execution and system outputs. The paper does not explicitly state whether the training data or the evaluation benchmarks include potentially harmful code (e.g., malware, exploits, or scripts for unauthorized access). Given the model’s ability to simulate “error handling” and “failure modes” (Section e002, Appendix), there is a risk that the model could learn to generate or simulate dangerous payloads if not strictly constrained. The “Rule-Based Verifier” should be explicitly described as including safety filters to prevent the generation of harmful content during the simulation phase.

Recommendations:

1. Add a “Safety and Limitations” section explicitly discussing dual-use risks, particularly regarding misinformation and social engineering.

2. Expand the discussion on LLM judge reliability and the potential for reward hacking to undermine safety evaluations.
3. Clarify the safety protocols for training data and evaluation benchmarks in sensitive domains like SWE and Terminal.

Action items.

- **[writing]** The paper describes training agents on ‘fictional-world construction’ (e.g., 2030 Mars migration) and ‘controllable simulation’ of errors. Explicitly address the dual-use risk of these capabilities being used to generate highly realistic disinformation, deepfakes, or social engineering scenarios. A dedicated ‘Safety and Limitations’ subsection is required.
- **[writing]** The evaluation relies heavily on LLM judges (e.g., GPT-5.2) to score ‘Realism’ and ‘Factuality’ of simulated environments. Discuss the risk of reward hacking or ‘self-praise’ (noted in Section e002) leading to over-optimistic safety assessments where the model learns to mimic safety signals without genuine alignment.
- **[writing]** The benchmark includes ‘SWE’ (Software Engineering) and ‘Terminal’ domains where the model predicts execution outputs. Clarify if the training data or evaluation includes potentially harmful code (e.g., malware, exploits) and confirm that the ‘Rule-Based Verifier’ includes safety filters to prevent the generation of dangerous payloads during the simulation phase.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a comprehensive framework for Language World Models (LWMs) with extensive quantitative results across seven domains. However, the strength of the scientific evidence regarding the superiority of Simulation-based Reinforcement Learning (Sim RL) over Real RL requires further statistical rigor.

In Section “Application I: Environment Simulator,” the authors claim that Sim RL achieves a higher F1 Item score (50.3%) compared to Real RL (45.6%) on the WideSearch benchmark (Table 3, Figure 4). This conclusion is drawn from a comparison of two specific model configurations (35B and 397B). Without reporting results from multiple random seeds or providing confidence intervals, it is impossible to determine if this ~4.7% gain is statistically significant or a result of random variance in the evaluation. Given the high variance often seen in LLM benchmarks, a single run comparison is insufficient to support the strong claim that “Sim RL surpasses Real RL.”

Additionally, the “Rule-Based Verification” results in Table 4 present a nuanced picture. While the proposed method (Qwen3.5-397B-A17B) achieves a high average score, the text states

it “surpasses frontier models on GUI domains.” However, the table shows GPT-5.4 achieving a higher average (72.93%) than the proposed method (67.12%). The claim of superiority appears to rely on specific sub-metrics (e.g., Android or Web) rather than the aggregate. The authors should explicitly define which specific GUI domains and metrics support this claim to avoid overgeneralization.

Finally, the qualitative analysis of reasoning patterns (Section “LWM Reasoning Patterns”) relies on 129 thinking traces. While the identification of “Wait!” interrupts is interesting, the sample size is relatively small for making broad claims about the model’s general reasoning capabilities across four diverse domains. The authors should clarify the distribution of these traces across domains (Terminal, MCP, Search, SWE) to ensure the observed patterns are not artifacts of a specific domain’s difficulty or data composition.

Action items.

- **[science]** The claim that Sim RL (50.3%) outperforms Real RL (45.6%) on WideSearch (Table 3, Fig 4) lacks statistical validation. With only two data points per condition (35B and 397B models), the difference could be noise. Report confidence intervals or p-values from multiple seeds to rule out random variance.
- **[science]** The ‘Rule-Based Verification’ results (Table 4) show GPT-5.4 outperforming the proposed method on average (72.93% vs 67.12%), yet the text claims the proposed method ‘surpasses frontier models on GUI domains’ without specifying which specific GUI metrics or domains justify this generalization. Clarify the scope of this claim or adjust the wording.
- **[science]** The analysis of ‘Deliberative Self-Correction’ cites 1,347 interrupts across 129 turns (approx. 10.4/turn). This sample size (n=129) is small for drawing broad conclusions about reasoning patterns across four distinct domains. Provide the distribution of turns per domain and confirm if the ‘Wait!’ pattern is statistically significant across all domains or driven by a single one (e.g., SWE).

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents extensive quantitative results across seven domains, but the statistical rigor supporting the central claims is insufficient for a paper of this scope.

First, the primary comparison in Section 3 (Experiments) and Table 1 relies entirely on point estimates. The authors claim \method-397B-A17B (58.71) surpasses GPT-5.4 (58.25) and other baselines. However, no measures of uncertainty (standard deviation, standard error, or 95% confidence intervals) are provided for these scores. In LLM benchmarking, performance can fluctuate significantly based on prompt variations or random seeds. Without reporting the variance across multiple runs or using bootstrap resampling to generate confidence intervals, the

statistical significance of a 0.46 point difference cannot be established. The claim of superiority is currently unsupported by inferential statistics.

Second, the paper conducts a large number of comparisons across 14 baselines and 7 domains (Section 3, Table 1; Section 4, Table 3). The authors highlight specific gains (e.g., +11.28 on Claw-Eval, +15.50 on BFCL v4) without addressing the multiple-comparisons problem. With this many hypothesis tests, the probability of false positives (Type I errors) is high. The manuscript should apply a correction method (e.g., Bonferroni, Holm-Bonferroni, or Benjamini-Hochberg FDR) to the reported p-values or explicitly state that the results are exploratory and not corrected for multiple testing.

Third, the qualitative analysis in Section 5 (Analysis) draws strong conclusions from a small, non-randomized sample. The authors analyze “129 thinking traces” to identify patterns like “Deliberative Self-Correction” (1,347 interrupts). The selection criteria for these 129 turns are not specified. If these were cherry-picked to demonstrate the model’s capabilities, the statistical validity of the generalization is compromised. A random sampling protocol or a power analysis justifying the sample size is necessary to support the claim that these patterns are representative of the model’s general behavior.

Finally, while the rule-based verification results in the Appendix (Tables 4-6) show clear accuracy percentages, the statistical methods used to aggregate these scores (e.g., weighted averages across domains) are not detailed. The weighting scheme for the “Avg.” column in Table 4 should be explicitly defined to ensure reproducibility.

To address these issues, the authors must re-run experiments to report variance, apply statistical significance testing with appropriate corrections, and clarify the sampling methodology for the qualitative analysis.

Action items.

- **[science]** Section 3 (Experiments) and Table 1 report point estimates (e.g., 58.71 vs 58.25) without confidence intervals, standard errors, or p-values. Given the high variance in LLM benchmarks, statistical significance testing (e.g., bootstrap or t-tests) is required to validate the claimed superiority over GPT-5.4.
- **[science]** Section 4 (Terminal-Bench 2.0) and Table 3 show improvements (e.g., +8.96 average) but lack multiple-comparisons correction. With 14 baselines and 7 domains tested, the risk of Type I error is high. Apply Bonferroni or FDR correction to reported significance claims.
- **[science]** Section 5 (Analysis) cites ‘129 thinking traces’ and ‘1,347 interrupts’ as evidence of reasoning patterns. The sampling strategy for these traces is not described (random vs. cherry-picked). A statistical power analysis or randomization protocol is needed to ensure these qualitative patterns are representative.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high level of technical sophistication, but the writing quality occasionally suffers from dense phrasing and inconsistent terminology that impedes immediate readability. While the core arguments are sound, several sections require polishing to meet the clarity standards of top-tier venues.

In the **Introduction**, the slogan “CPT injects, SFT activates, RL sharpens” is stylistically distinct but grammatically abrupt. It interrupts the flow of the narrative. A more explicit phrasing, such as “CPT injects world knowledge, SFT activates reasoning capabilities, and RL sharpens simulation fidelity,” would better integrate the concept into the prose without sacrificing brevity.

Section 2 (Preliminaries) introduces the unified schema using the symbol $\oplus$ (trajectory = `system_prompt  $\oplus$  [turn1, ..., turnT]`). This operator is not defined in the text or the notation table. Readers must infer that it represents concatenation. Explicitly defining this symbol or using standard notation (e.g., $\oplus$ as concatenation) is necessary to avoid confusion.

Section 5 (Experiments) and **Section 6** exhibit inconsistent naming conventions for benchmarks. The text alternates between “SWE-B V” and “SWE-Bench Verified,” and “Claw” versus “Claw-Eval.” While these may be shorthand, consistency is vital for professional presentation. The full names used in Table 1 should be adopted throughout the main text, with abbreviations defined once.

Finally, in **Section 3 (Training Recipe)**, the description of the reward function (“Reward = 90% Five-Dimensional Rubric...”) is slightly ambiguous. It is unclear if this refers to the weight of the components in the loss function or the composition of the final score. Rephrasing to “The reward signal is a weighted sum: 90% from the Five-Dimensional Rubric...” would eliminate this potential ambiguity.

Overall, the paper is well-structured, but these specific edits will significantly enhance the reader’s ability to follow the technical narrative without stumbling over undefined symbols or inconsistent terminology.

Action items.

- **[writing]** In Section 2 (Preliminaries), the definition of the unified schema uses the symbol `\oplus` without prior definition or context. Define this operator or replace with standard concatenation notation to ensure clarity for readers unfamiliar with the specific formalism.
- **[writing]** Section 5 (Experiments) and Section 6 (Terminal-Bench 2.0) contain inconsistent capitalization and spacing in benchmark names (e.g., ‘SWE-B V’ vs ‘SWE-Bench Verified’, ‘Claw’ vs ‘Claw-Eval’). Standardize these names to match the full titles used in the Introduction and Table 1 for consistency.
- **[writing]** The phrase ‘CPT injects, SFT activates, RL sharpens’ in the Introduction is a catchy slogan but lacks immediate grammatical parallelism with the surrounding prose. Consider rephrasing to ‘CPT injects knowledge, SFT activates reasoning, and RL sharpens fidelity’ or similar to improve flow and clarity.

- **[writing]** In Section 3 (Training Recipe), the sentence ‘Reward = 90% Five-Dimensional Rubric (LLM Judge) + 10% Rule-Based Verifier’ is slightly ambiguous regarding whether the percentages refer to weight or score contribution. Rephrase to ‘The reward signal comprises a weighted sum: 90% from the Five-Dimensional Rubric... and 10% from the Rule-Based Verifier’.